Green MIMO Detection

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1. ABSTRACT

Background

- MIMO systems are widely applied in modern communication technologies.

Challenge

- The computational complexity of detection in MIMO systems is increasing, leading to higher hardware burden and latency.

Proposed Method

- Green MIMO Detection approach:

  a low-complexity K-Best detector is presented for MIMO systems.

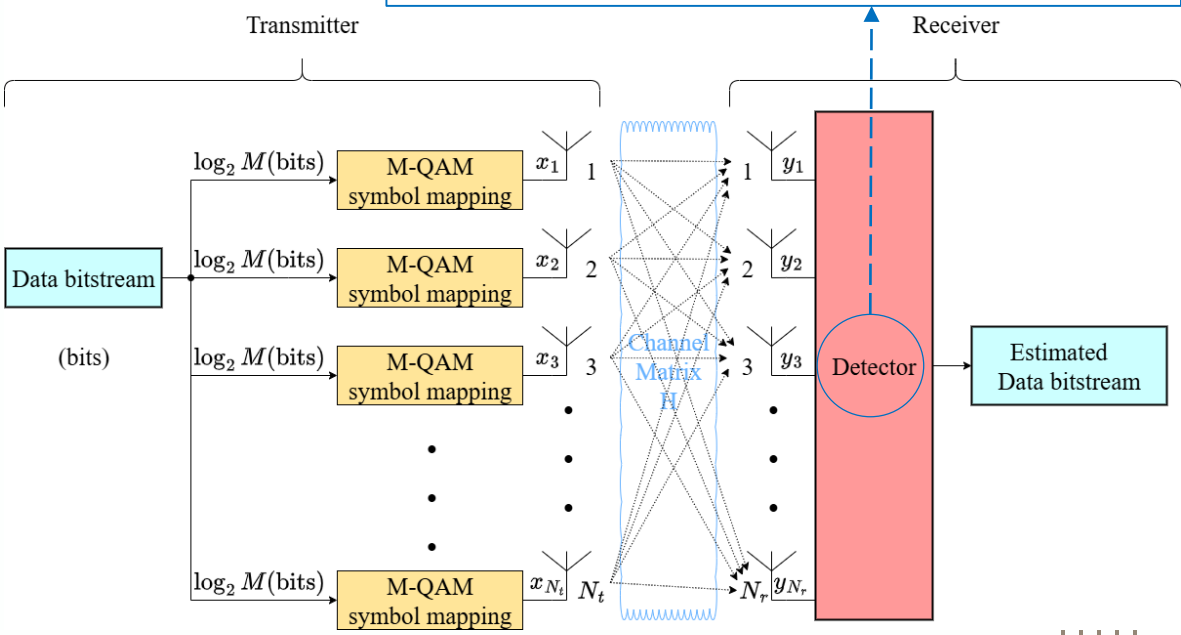
2. MIMO SYSTEM MODEL

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n}$$

where

- $\boldsymbol{y}$: received vector $(N_r \times 1)$
- $\boldsymbol{H}$: channel matrix $(N_r \times N_t)$
- $\boldsymbol{x}$: transmitted symbol vector $(N_t \times 1)$
- $\boldsymbol{n}$: complex AWGN vector $(N_r \times 1)$

Detect $\boldsymbol{x}$ from the received vector $\boldsymbol{y}$ and the channel matrix $\boldsymbol{H}$



3. PROPOSED ALGORITHM: GREEN MIMO DETECTION

- Green MIMO Detection is modified from MLD and Traditional K-Best detector.

A. Maximum likelihood detector (MLD): [1]

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n}$$

$$\hat{\boldsymbol{x}}_{MLD} = \arg \min_{\boldsymbol{x} \in \boldsymbol{C}^{N_t}} \|\boldsymbol{y} - \boldsymbol{H}\boldsymbol{x}\|^2$$

For example, assume $N_r = N_t = 2$ with 4-QAM. Then $\boldsymbol{x}$ is a 2×1 vector, where $\boldsymbol{x}_j \in \{ 1 + i, 1 - i, -1 + i, -1 - i\}$.

$\boldsymbol{x} = \begin{bmatrix} 1+i \\ 1+i \end{bmatrix}$, or $\begin{bmatrix} -1+i \\ 1+i \end{bmatrix}$, or $\begin{bmatrix} 1-i \\ -1-i \end{bmatrix}$, ..., total 16 possible symbol vectors.

[1] Y. S. Cho, J. Kim, W. Y. Yang, and C.-G. Kang, "MIMO-OFDM Wireless Communications with MATLAB," Wiley, Hoboken, NJ, USA, 2010.

B. Traditional K-Best detector: [2]

$$\begin{aligned} \boldsymbol{y} &= \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n} = \boldsymbol{Q}\boldsymbol{R}\boldsymbol{x} + \boldsymbol{n} \\ \Rightarrow \quad \boldsymbol{Q}^H\boldsymbol{y} &= \boldsymbol{Q}^H\boldsymbol{Q}\boldsymbol{R}\boldsymbol{x} + \boldsymbol{Q}^H\boldsymbol{n} \\ \Rightarrow \quad \tilde{\boldsymbol{y}} &= \boldsymbol{R}\boldsymbol{x} + \tilde{\boldsymbol{n}} \end{aligned}$$

$$\hat{\boldsymbol{x}} = \arg \min_{\boldsymbol{x} \in \boldsymbol{C}^{N_t}} \|\tilde{\boldsymbol{y}} - \boldsymbol{R}\boldsymbol{x}\|^2 = \arg \min_{\boldsymbol{x} \in \boldsymbol{C}^{N_t}} \sum_{i=1}^{N_t} \left| \tilde{\boldsymbol{y}}_i - \boldsymbol{R}_{i,i}\boldsymbol{x}_i - \sum_{j=i+1}^{N_t} \boldsymbol{R}_{i,j}\boldsymbol{x}_j \right|^2$$

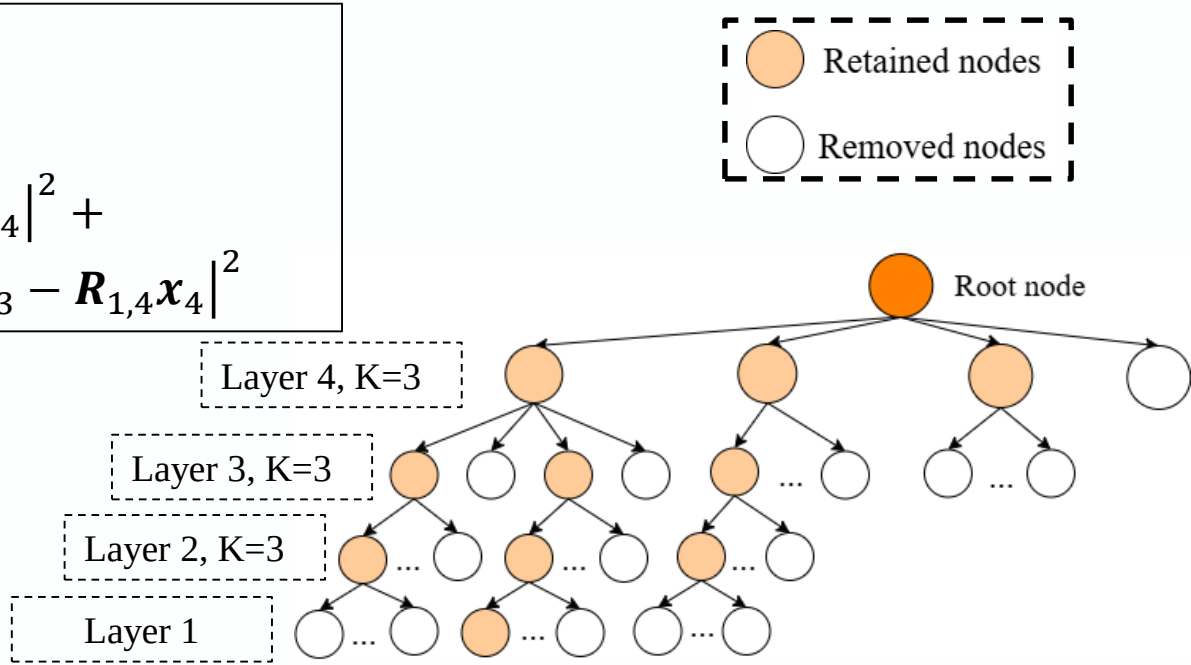
[2] Y.-X. Liu, S.-J. Jihang, and Y.-L. Ueng, “An Efficient K-Best MIMO Detector for Large Modulation Constellations,” in IEEE Open Journal of Circuits and Systems, vol. 5, pp. 2-16, 2024.

EXAMPLE: $\hat{\boldsymbol{x}} = \arg\min_{\boldsymbol{x}\in \boldsymbol{C}^{N_t}} \|\tilde{\boldsymbol{y}} - \boldsymbol{R}\boldsymbol{x}\|^2$, let $N_r = N_t = 4$ with 4-QAM.

$$\tilde{\boldsymbol{y}} = \begin{bmatrix} \tilde{\boldsymbol{y}}_1 \\ \tilde{\boldsymbol{y}}_2 \\ \tilde{\boldsymbol{y}}_3 \\ \tilde{\boldsymbol{y}}_4 \end{bmatrix}, \quad \boldsymbol{R} = \begin{bmatrix} \boldsymbol{R}_{1,1} & \boldsymbol{R}_{1,2} & \boldsymbol{R}_{1,3} & \boldsymbol{R}_{1,4} \\ 0 & \boldsymbol{R}_{2,2} & \boldsymbol{R}_{2,3} & \boldsymbol{R}_{2,4} \\ 0 & 0 & \boldsymbol{R}_{3,3} & \boldsymbol{R}_{3,4} \\ 0 & 0 & 0 & \boldsymbol{R}_{4,4} \end{bmatrix}, \quad \boldsymbol{x} = \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \boldsymbol{x}_3 \\ \boldsymbol{x}_4 \end{bmatrix}$$

4-QAM:
$\{1 + i, 1 - i, -1 + i, -1 - i\}$

$$\begin{aligned} \eta = & \left|\tilde{\boldsymbol{y}}_4 - \boldsymbol{R}_{4,4}\boldsymbol{x}_4\right|^2 + \\ & \left|\tilde{\boldsymbol{y}}_3 - \boldsymbol{R}_{3,3}\boldsymbol{x}_3 - \boldsymbol{R}_{3,4}\boldsymbol{x}_4\right|^2 + \\ & \left|\tilde{\boldsymbol{y}}_2 - \boldsymbol{R}_{2,2}\boldsymbol{x}_2 - \boldsymbol{R}_{2,3}\boldsymbol{x}_3 - \boldsymbol{R}_{2,4}\boldsymbol{x}_4\right|^2 + \\ & \left|\tilde{\boldsymbol{y}}_1 - \boldsymbol{R}_{1,1}\boldsymbol{x}_1 - \boldsymbol{R}_{1,2}\boldsymbol{x}_2 - \boldsymbol{R}_{1,3}\boldsymbol{x}_3 - \boldsymbol{R}_{1,4}\boldsymbol{x}_4\right|^2 \end{aligned}$$



The number of retained nodes at each layer must be $\leq$ K

C. Green MIMO Detection

$$\boldsymbol{y} = \boldsymbol{H}\boldsymbol{x} + \boldsymbol{n}$$

Real-Valued Decomposition (RVD)

$$\begin{bmatrix} \Re(\boldsymbol{y}) \\ \Im(\boldsymbol{y}) \end{bmatrix}_{2N_r \times 1} = \begin{bmatrix} \Re(\boldsymbol{H}) & -\Im(\boldsymbol{H}) \\ \Im(\boldsymbol{H}) & \Re(\boldsymbol{H}) \end{bmatrix}_{2N_r \times 2N_t} \begin{bmatrix} \Re(\boldsymbol{x}) \\ \Im(\boldsymbol{x}) \end{bmatrix}_{2N_t \times 1} + \begin{bmatrix} \Re(\boldsymbol{n}) \\ \Im(\boldsymbol{n}) \end{bmatrix}_{2N_r \times 1}$$

$$\Rightarrow \quad \boldsymbol{y_r} = \boldsymbol{H_r}\boldsymbol{x_r} + \boldsymbol{n_r} \text{ (real-valued system model)}$$

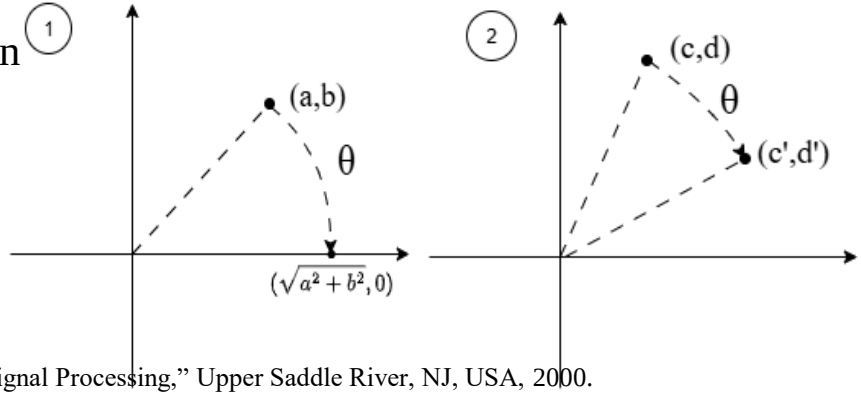
$$\boldsymbol{y_r} = \boldsymbol{H_r}\boldsymbol{x_r} + \boldsymbol{n_r} = \boldsymbol{Q}\boldsymbol{R}\boldsymbol{x_r} + \boldsymbol{n_r}$$

$$\Rightarrow \quad \boldsymbol{Q}^H\boldsymbol{y_r} = \boldsymbol{Q}^H\boldsymbol{Q}\boldsymbol{R}\boldsymbol{x_r} + \boldsymbol{Q}^H\boldsymbol{n_r}$$

$$\Rightarrow \quad \tilde{\boldsymbol{y}} = \boldsymbol{R}\boldsymbol{x_r} + \tilde{\boldsymbol{n}}$$

- Given rotations [3] are applied to QR decomposition

$$H = \begin{bmatrix} a & c \\ b & d \end{bmatrix} \xrightarrow{\text{Given rotations}} R = \begin{bmatrix} \sqrt{a^2+b^2} & c' \\ 0 & d' \end{bmatrix}$$



[3] T. K. Moon and W. C. Stirling, "Mathematical Methods and Algorithms for Signal Processing," Upper Saddle River, NJ, USA, 2000.

- $SNR_i = (E\{|x_i|^2\} \times |\boldsymbol{R}_{i,i}|^2)/E\{|\tilde{n}_i|^2\} \sim |\boldsymbol{R}_{i,i}|^2$ [4]

$$\boldsymbol{H_r} = \begin{bmatrix} \boldsymbol{H_r}_{1,1} & \boldsymbol{H_r}_{1,2} \\ \boldsymbol{H_r}_{2,1} & \boldsymbol{H_r}_{2,2} \end{bmatrix} \Rightarrow \boldsymbol{R} = \begin{bmatrix} \boldsymbol{R}_{1,1} & \boldsymbol{R}_{1,2} \\ \boldsymbol{0} & \boldsymbol{R}_{2,2} \end{bmatrix}, \boldsymbol{R}_{1,1} = \left\| \boldsymbol{H_r}_{:,1} \right\| = \sqrt{\boldsymbol{H_r}_{1,1}{}^2 + \boldsymbol{H_r}_{2,1}{}^2}$$

$$, \boldsymbol{R}_{2,2} = \left\| \boldsymbol{H_r}_{:,2} \right\| \sin\theta = \boldsymbol{Q}_{:,2}^T \boldsymbol{H_r}_{:,2}$$

where

- $\boldsymbol{H_r}_{:,1}$: the first column of $\boldsymbol{H_r}$
- $\boldsymbol{H_r}_{:,2}$: the second column of $\boldsymbol{H_r}$
- θ: the angle between $\boldsymbol{H_r}_{:,1}$ and $\boldsymbol{H_r}_{:,2}$
- $\boldsymbol{Q}_{:,2}$: the second column of $\boldsymbol{Q}$

[4] Li Liu, Jin. Wang, Dong. Yan, Fu. Liu, and Xin Song, “Sorted QR Decomposition Based Tree Search Detection Algorithm for MIMO system,” in Proc. 2012 8th Int. Conf. on Wireless Commun., Networking and Mobile Computing, Sep. 2012, pp. 1–4.

4. BER PERFORMANCE

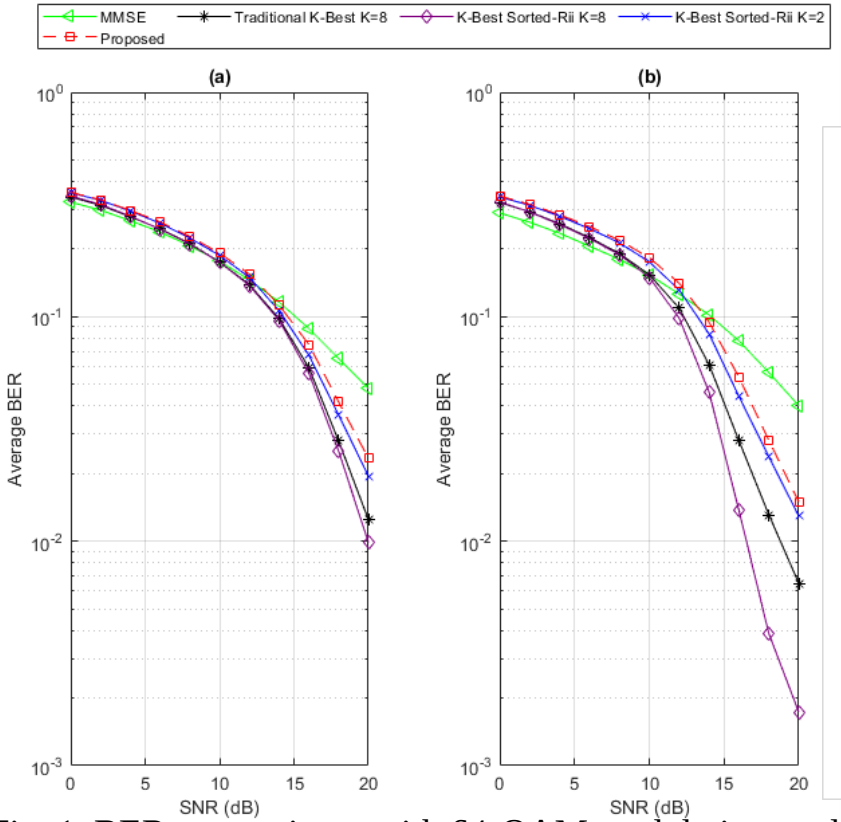


Fig. 1. BER comparisons with 64-QAM modulation, and (a) $N_t = N_r = 4$, and (b) $N_t = N_r = 8$.

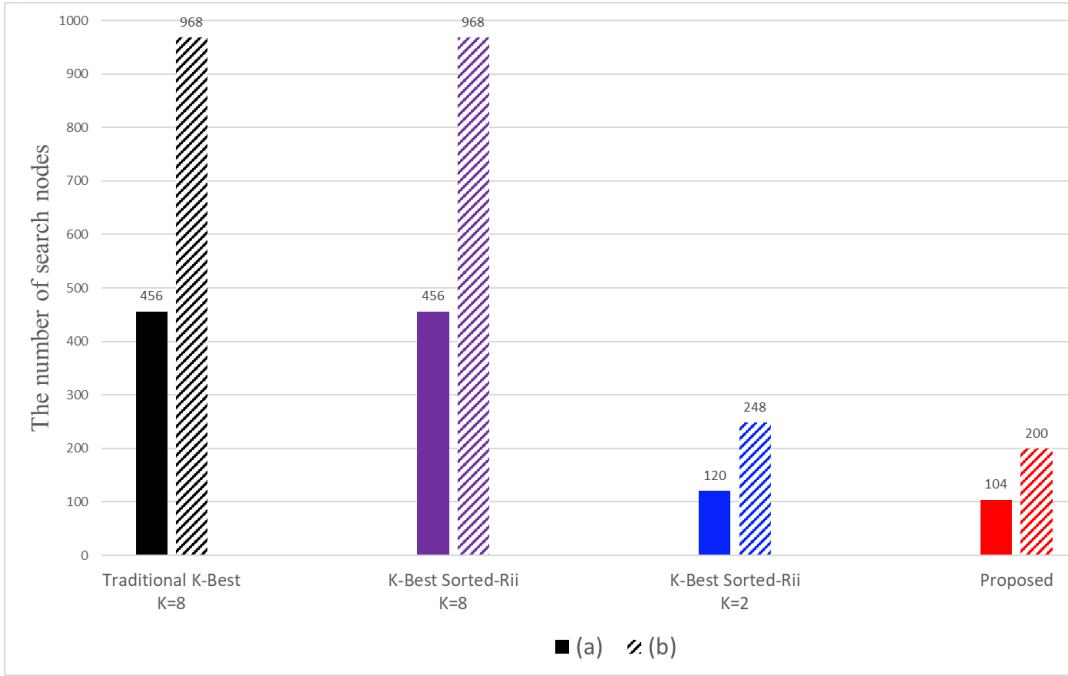


Fig. 2. The number of search nodes with 64-QAM modulation, and (a) $N_t = N_r = 4$, and (b) $N_t = N_r = 8$.

5. CONCLUSION

- Simulation and complexity results demonstrate that Green MIMO detection can achieve competitive performance with modest computational complexity.

- This proposed algorithm is well-suited for hardware implementation.

Thank you for your attention